

Conjecture 00000003888 is false:
the Hilbert–Kunz density function of $(k[x, y, z], (x, y, z))$
is $x^2/2$ on $[0, 1)$, hence not piecewise linear

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1 The conjecture and the reading

Statement (English). *Definition: The Hilbert–Kunz density function is the limit of relative eHK as a function of parameter x . Conjecture: The HK density function of a fixed ideal is piecewise linear with finitely many breakpoints; conversely, there exist explicit one-parameter ideal families whose breakpoint counts grow unboundedly.* The Chinese text says the same: the HK density function of a fixed ideal is piecewise linear with finitely many breakpoints, and conversely there is an explicit one-parameter family of ideals whose breakpoint counts grow without bound.

Reading. The conjecture is a conjunction (Clause 1) $\wedge$ (Clause 2). We refute Clause 1, which refutes the conjunction whatever Clause 2 means. We do not address Clause 2. The text does not restrict the ring or its dimension, so Clause 1 says: for every pair (R, I) for which the HK density function is defined, $\text{HKd}(R, I)$ is piecewise linear with finitely many breakpoints. We exhibit one pair, $R = k[x, y, z]$ with the standard grading and $I = (x, y, z)$, over any field k of characteristic $p > 0$, for which this fails. For curves (standard graded rings of dimension 2), piecewise linearity is a theorem of Trivedi (quoted below). Our counterexample has dimension 3, which the text does not exclude.

2 Definitions

Source. V. Trivedi, *Hilbert–Kunz density function and Hilbert–Kunz multiplicity*, <https://arxiv.org/abs/1510.03294>. The quotes below were retrieved from the arXiv abstract and from the ar5iv HTML of the paper (<https://ar5iv.labs.arxiv.org/html/1510.03294>).

- Abstract: “For a pair (M, I) , where M is finitely generated graded module over a standard graded ring R of dimension d , and I is a graded ideal with $\ell(R/I) < \infty$, we introduce a new invariant $\text{HKd}(M, I)$ called the Hilbert–Kunz density function.”
- Introduction: “ $\text{HKd}(M, I)(x) = f(x) = \lim_{n \rightarrow \infty} g_n(x) = \lim_{n \rightarrow \infty} f_n(x)$, where $f_n(x) = (1/q^{d-1})\ell(M/I^{[q]}M)_{[xq]}$ ”. Notations 2.1: “Fix $n \in \mathbb{N}$ and denote $q = p^n$ ”.
- Frobenius power: “ $I^{[p^n]}$ = n -th Frobenius power of I = the ideal generated by p^n -th powers of elements of I ”.
- Dimension 2: “the density function $\text{HKd}(R, \mathfrak{m})$, is a piecewise linear polynomial with rational coefficients”.

Definition 1 (Lean: `frobeniusPower`, `gradedPiece`, `hkApprox`, `hkDensity`). Let k be a field of characteristic $p > 0$, $R = k[x_\sigma]$ with $|\sigma| = d$ and the standard grading, and I an ideal. $I^{[q]}$ is the ideal generated by $\{f^q : f \in I\}$. The degree- m piece $(R/J)_m$ is the image of the degree- m homogeneous polynomials in R/J , and $\ell((R/J)_m) = \dim_k(R/J)_m$; it is 0 for $m < 0$. Then

$$f_n(x) = \frac{1}{q^{d-1}} \dim_k(R/I^{[q]})_{\lfloor xq \rfloor}, \quad q = p^n, \quad \text{HKd}(R, I)(x) = \lim_{n \rightarrow \infty} f_n(x).$$

In Lean, $p = \text{ringChar } k$, $d = \text{Nat.card } \sigma$ (and `ringKrullDim_eq` proves $\dim k[x_\sigma] = |\sigma|$), and `HKd` is `limUnder atTop`, which equals the limit wherever it exists. This is Trivedi's definition for $M = R$. A module of length ℓ over R concentrated in one degree is a $k = R_0$ -vector space of dimension ℓ .

Definition 2 (Lean: `IsPiecewiseLinearOn`, `IsPiecewiseLinearPartition`). $f : \mathbb{R} \rightarrow \mathbb{R}$ is *piecewise linear with finitely many breakpoints on s* if there is a finite set B such that f is affine ($f(x) = \mu x + c$) on every open interval contained in $s \setminus B$. Values at breakpoints are unconstrained and continuity is not required. The partition form says there are $a = t_0 < t_1 < \dots < t_k = b$ with f affine on every $[t_i, t_{i+1}]$. Lemma `IsPiecewiseLinearPartition.isPiecewiseLinearOn` proves that the partition form on $[a, b]$ implies `IsPiecewiseLinearOn` on $[a, b]$, with $B = \{t_0, \dots, t_k\}$. “Linear” pieces are read as affine, which is the weaker condition. So the refutation also covers the reading with pieces of the form μx .

3 The proof

Fix a field k of characteristic $p > 0$, $R = k[x, y, z]$ ($d = 3$) and $\mathfrak{m} = (x, y, z)$ (Lean: `maxIdeal`).

Lemma 3 (`maxIdeal_isHomogeneous`, `finrank_quotient_maxIdeal`). $\mathfrak{m}$ is homogeneous, and $\dim_k R/\mathfrak{m} = 1$. So $(R, \mathfrak{m})$ is in the scope of the definition.

Proof. It is generated by the homogeneous elements x, y, z . The kernel of $f \mapsto f(0)$ is $\mathfrak{m}$: a polynomial lies in the monomial ideal $\mathfrak{m}$ iff every monomial in its support is divisible by some variable, that is, iff its constant coefficient is 0. So $R/\mathfrak{m} \cong k$. $\square$

Lemma 4 (`frobeniusPower_maxIdeal`). For $q = p^n$, $\mathfrak{m}^{[q]} = (x^q, y^q, z^q)$.

Proof. $\supseteq$ is clear. For $\subseteq$, induct on $f \in \mathfrak{m} = \text{span}(x, y, z)$. The generators satisfy $x_i^q \in (x^q, y^q, z^q)$, and $0^q = 0$. In characteristic p , $(a + b)^q = a^q + b^q$ (`add_pow_char_pow`), and $(ra)^q = r^q a^q$. $\square$

Lemma 5 (`hilbertFunction_monomialIdeal`, `hilbertFunction_Gq`). For $0 \leq m < q$, $\dim_k(R/(x^q, y^q, z^q))_m = \binom{m+2}{2}$.

Proof. For a monomial ideal, $\dim_k(R/J)_m$ is the number of degree- m monomials not divisible by any generator. This is proved in Lean via `mem_ideal_span_monomial_image`, reusing the accepted solution of conjecture 00000000539. If $m < q$, no degree- m monomial is divisible by x_i^q . So all $\binom{m+2}{2}$ degree- m monomials are standard (stars and bars, `card_finsuppAntidiag_nat_eq_choose`). $\square$

Theorem 6 (`tendsto_hkApprox`, `hkDensity_maxIdeal`). For $0 \leq x < 1$, $f_n(x) \rightarrow x^2/2$ as $n \rightarrow \infty$. Hence $\text{HKd}(R, \mathfrak{m})(x) = x^2/2$ on $[0, 1)$.

Proof. Let $q = p^n$ and $m = \lfloor xq \rfloor$. Then $0 \leq m \leq xq < q$. By the lemmas, $f_n(x) = \binom{m+2}{2}/q^2 = \frac{1}{2}(m/q + 2/q)(m/q + 1/q)$. Since $x - 1/q < m/q \leq x$ and $q \rightarrow \infty$, we get $m/q \rightarrow x$ and $1/q \rightarrow 0$. $\square$

This agrees with Trivedi's Example 3.1 (retrieved), stated for $R = k[X_0, \dots, X_d]$ and its graded maximal ideal: " $HKd(\mathbb{P}_k^d)(x) = x^d/d!$, for $0 \leq x < 1$ ". Here $d = 2$. Our Lean proof does not use this example.

Theorem 7 (`not_affine`, `not_isPiecewiseLinearOn`). *If $f = x^2/2$ on $(0, 1)$, then f is not affine on any open interval $(a, b) \subseteq (0, 1)$ with $a < b$. Hence f is not piecewise linear with finitely many breakpoints on any $s \supseteq (0, 1)$.*

Proof. Let $u = (3a+b)/4$, $v = (a+b)/2$ and $w = (a+3b)/4$. An affine f has $f(u) + f(w) - 2f(v) = 0$, but here this quantity is $(b-a)^2/16 > 0$. Given a finite B , pick $x \in (0, 1) \setminus B$. The set $(0, 1) \setminus B$ is open, so it contains an interval $(x - \varepsilon, x + \varepsilon)$, on which f would have to be affine. $\square$

Theorem 8 (`not_clause1`, `conjecture_false`). *Let k be a field of characteristic $p > 0$. Clause 1, restricted to the polynomial rings $k[x_1, \dots, x_n]$ and their homogeneous ideals of finite colength, is false: $HKd(k[x, y, z], \mathfrak{m})$ is not piecewise linear on $\mathbb{R}$. Hence $\neg(\text{Clause 1} \wedge Q)$ for every proposition Q .*

Further variants are also proved. `hkDensity_not_isPiecewiseLinearOn` works on any domain $s \supseteq (0, 1)$, such as $[0, \infty)$ or $[0, 3]$. `hkDensity_not_isPiecewiseLinearPartition` covers the partition form on any $[a, b] \supseteq [0, 1]$. `limit_not_isPiecewiseLinearOn` applies to every function that is the pointwise limit of f_n on $(0, 1)$, so it does not depend on the `limUnder` convention.

4 Lean formalization

File `lean/Conjecture3888/Basic.lean` (Lean 4, Mathlib v4.33.1, only `import Mathlib`). Main theorem:

```
theorem conjecture_false (K : Type*) [Field K] (p : N) [Fact p.Prime]
  [CharP K p] (Clause2 : Prop) : not (Clause1 K /\ Clause2)
-- Clause1 K : for all n and I : Ideal (MvPolynomial (Fin n) K),
--   I homogeneous -> Module.Finite K (R / I) ->
--   IsPiecewiseLinearOn (hkDensity I) Set.univ
```

Axiom audit. `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound` for `conjecture_false`, `not_clause1`, `hkDensity_not_isPiecewiseLinearOn`, `hkDensity_not_isPiecewiseLinearPartition` and `limit_not_isPiecewiseLinearOn`. The file contains no `sorry` and no `native_decide`.